JESA

TECHNICAL DATA SHEET
ProLine PET-CF Filament

Product Overview

ProLine PET-CF is a modified PET-based 3D printing filament with 10% carbon-fiber reinforcement. It combines high temperature resistance, low shrinkage / low warping and easy printability, and can be processed on FDM printers without a heated chamber. Printed parts offer high rigidity and tensile strength and are suitable for long-term use at temperatures of up to 120°C.

Key Features

- Rigid, high-strength parts Low shrinkage / low warping for dimensionally
- stable prints Performs reliably at elevated temperatures (continuous use
- up to 120°C) Easy to print on non-heated chamber FDM printers
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Material Properties

PhysicalProperties			
Property	Test Method	Unit	Value
Density	ISO 1183	g/cm³	1.30–1.31
MFR (190°C / 2.16 kg)	ISO 1133	g/10 min	3–6
Moisture Absorption (23°C / 24 h)	ISO 62	%	< 0.3
Mechanical Properties			
Tensile strength (X-Y)	ISO 527	MPa %	75–85 4–5
Elongation at break (X-Y)	ISO 527	% MPa	1.5–2.5
Elongation at break (X-Z)	ISO 527	MPa	4500–5000
Flexural modulus (X-Y)	ISO 527	MPa	2000–2500
Flexural modulus (X-Z)	ISO 527	kJ/m²	140–150 25-
Flexural strength (X-Y)	ISO 178		30
Impact strength (X-Y)	ISO 180		
Thermodynamic Properties			
HDT @ 0.455 MPa (66 psi)	ISO 75	°C	190
Continuous Use Temperature	IEC 60216	°C	120

Recommended Printing Parameters

Parameter	Setting
Print Temperature	280-300°C
Bed Temperature	80–100°C
Bed Materials	Tempered glass, BuildTak, Carbon fiber board
Nozzle Diameter	ø 0.4 / 0.6 mm
Nozzle and gear material	High strength steel
Model Cooling Fan	OFF

Drying & Handling Notes

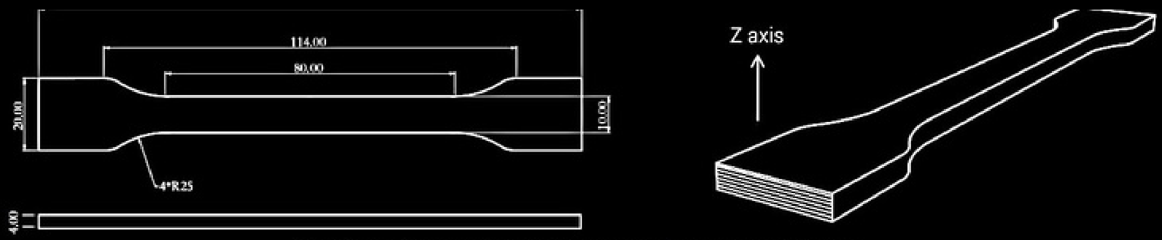
- Pre-dry: 90-100°C, minimum 12 h (hot-air oven) to reduce moisture-related printing issues and improve surface/part quality

Disclaimer

Test Specimens

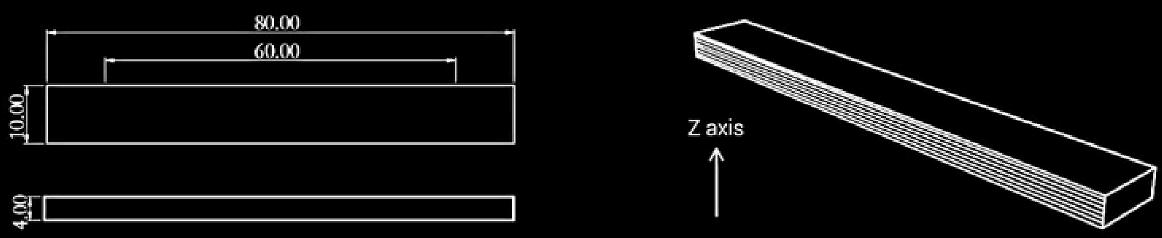
Test sample dimensions and printing direction

Figure 1. Tensile testing specimen; ASTM D638 (ISO 527, GB/T 1040)

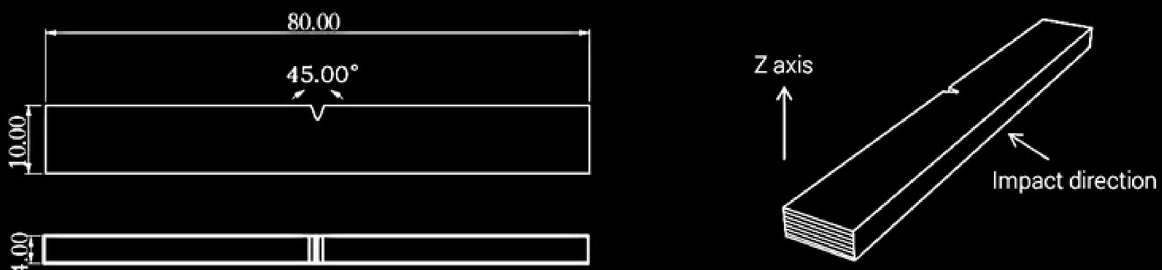


Tensile testing specimen; ASTM D638 (ISO 527, GB/T 1040)

Figure 2. Flexural and impact testing specimens; ASTM D790 (ISO 178, GB/T 9341) and ASTM D256 (ISO 179, GB/T 1043)



Flexural testing specimen; ASTMD790 (ISO 178, GB/T 9341)



Impact testing specimen; ASTM D256 (ISO 179, GB/T 1043)